

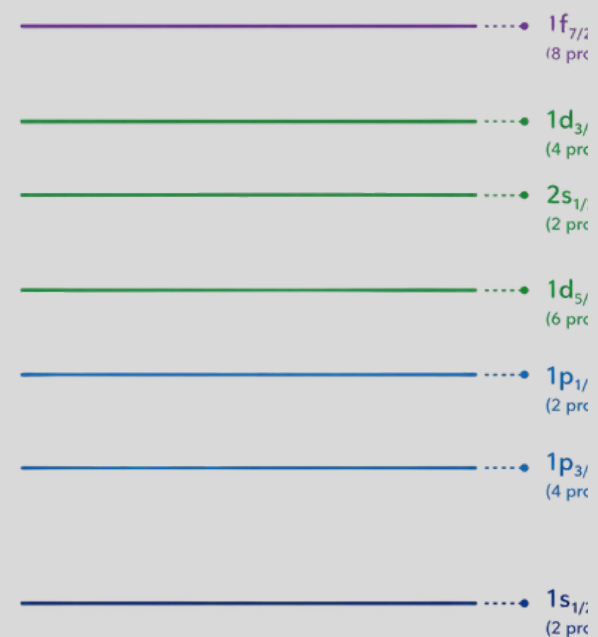
Physics-Informed Recovery of Nuclear Mean-Field Parameters

INPUTS

Sparse single particle spectra

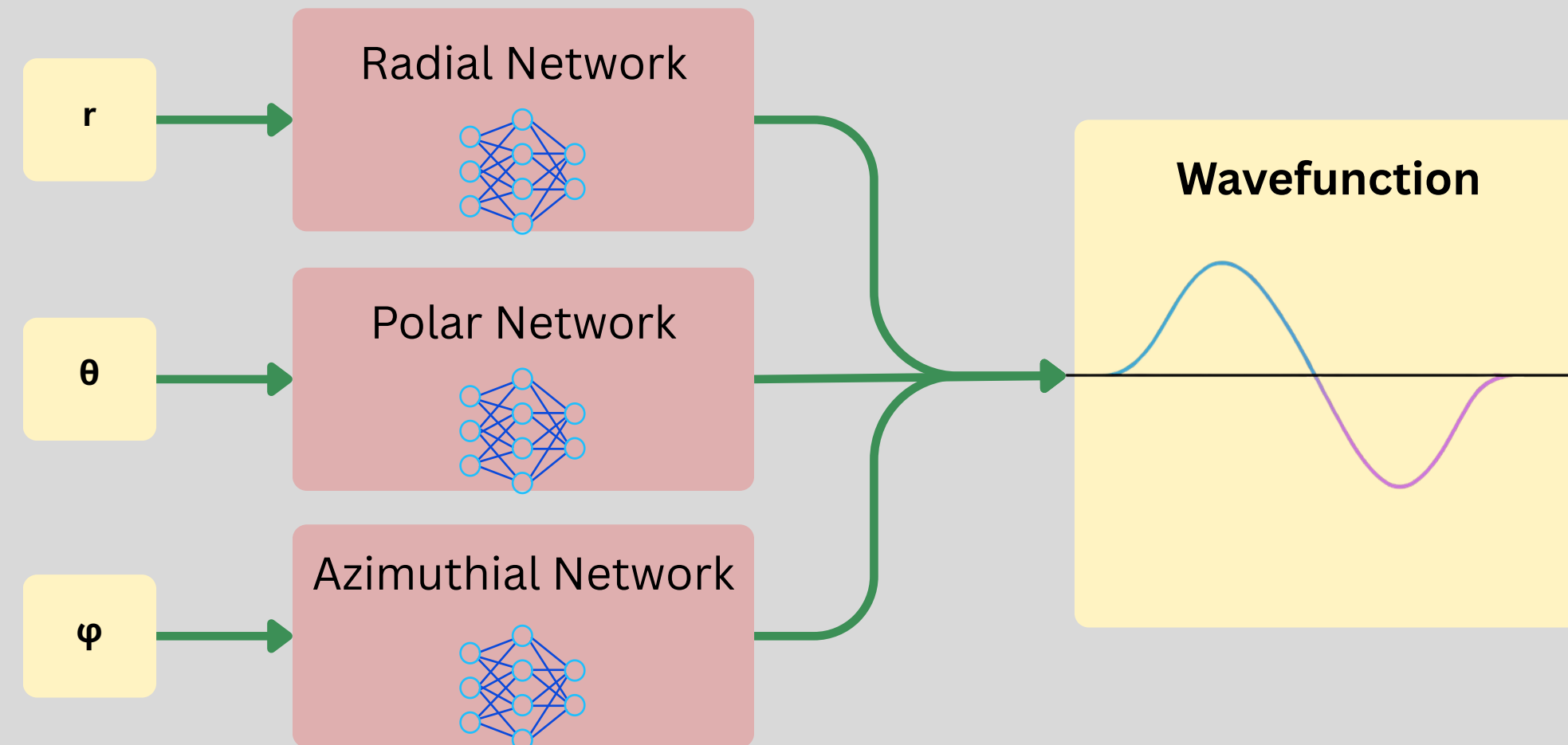
Energies, nuclear descriptors, quantum numbers

⁴⁰Ca Nuclear Energy Levels



⁴⁰Ca has 20 protons and 20 neutrons (closed shell)

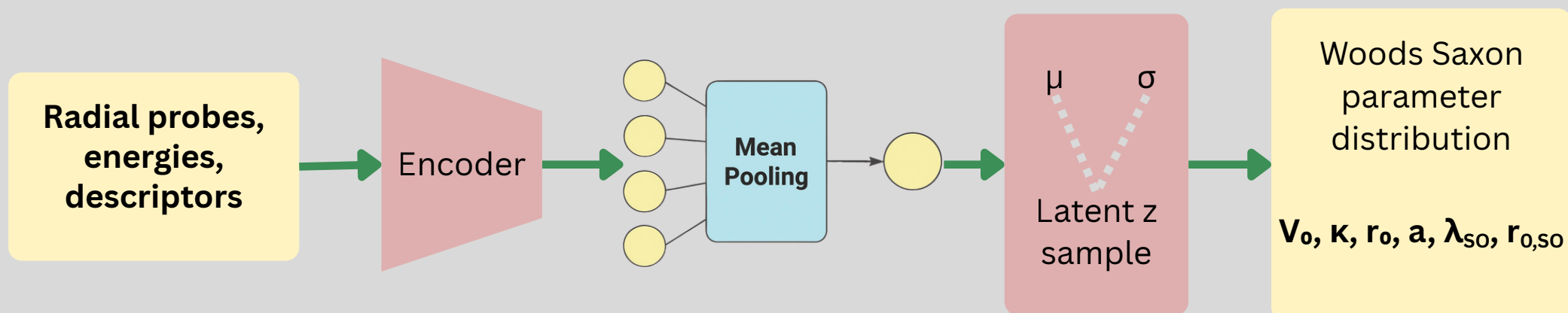
1. WaveNet (Forward Solver)



Physics Informed Objective

- Energy consistency
- Schrödinger residuals
- BCs
- Orthogonality
- Spin-Orbit Splitting
- KL divergence

2. Probabilistic ParamNet (Inverse Solver)

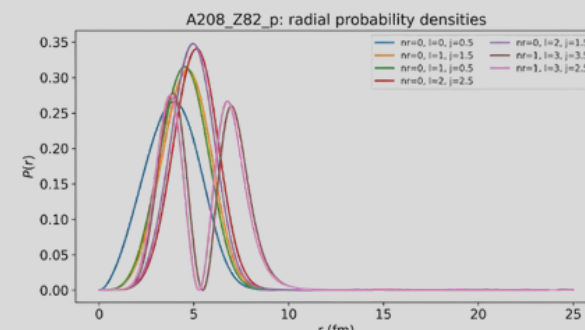


First: Forward Physics

Second: Inverse parameter update

Key results

- < 0.6% 6-parameter synthetic recovery.
- 0.8068 MeV Experimental spectral MAE (42 energy levels) compared to 0.7969 MeV MAE of reference parameterization.
- Wavefunctions representations.



(a) Radial probability densities $P(r)$ for selected proton states.